

Tracking Control for Robotic Manipulator Based on FABRIK Algorithm

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Abstract— This paper presents a tracking control problem for robotic manipulator based on the FABRIK inverse kinematics technique. The inverse kinematics uses the Kinematics equation to estimate the parameters of each degree of freedom in the manipulator. Recently, a new heuristic iterative approach has been introduced to solve the problem of Inverse Kinematics called FABRIK method. Our controller system combines the features of FABRIK, and the PID controller to achieve as simple, fast, and robust controlling system. The simulation results prove the effectiveness of the proposed technique. A demonstration of the results is included at the end of the paper.

Keywords— *Serial link manipulator, Tracking control, FABRIK, inverse Kinematics*

I. INTRODUCTION

The massive use of robotics manipulators in crucial industrial tasks such as welding, and painting has made controlling these arms a top research area in the past few years [1-4]. Moreover, the non-linearity nature of the robotic arms systems along with the time varying dynamic nature of the robotic arms systems, have also raised the system's complexity, and made the control task a more challenging mission. Another challenge manifests in producing realistic and plausible free-obstacle paths. Path generation algorithms must efficiently calculate the set of appropriate joint configurations for which the end effector navigates to the desired positions as smoothly and accuracy as possible.

The inverse kinematics is known as the problem of finding the desired positions for all joints knowing the desired position of the end-effector. Inverse kinematics can be used in the process of converting the position of the end effector from the Cartesian space to the positions of all joints in the joint space. Numerical inverse kinematics algorithms can fall into two main categories, namely Jacobian methods, and Newton methods [5].

A popular numerical method to compute the inverse kinematics is the Jacobian matrix. The Jacobian solution models the end-effector movement with respect to the instantaneous changes in the manipulator configurations represented by link translations, and joints angles. In [6], an inverse kinematic approach based on the Jacobian method is proposed. Authors assumed the endogenous configuration space tactic and invent a repeatable inverse kinematics technique for mobile manipulators. Jacobian methods can provide smooth postures, but at the expense of the high computational cost, due to singularity problems, and the complex matrix calculations.

The other numerical methods to perform the inverse kinematics problem for manipulators are the newton based methods. As Jacobian methods, these methods suffer also from complexity, and the high computational cost.

To overcome the complexity, and the high computational cost in both Jacobian and newton based methods, the FABRIK algorithm is proposed in [5]. FABRIK is a new method that is solving the problem of inverse kinematics. Simply, it uses the previous positions or parameters of each degree of freedom to find parameters in a mode called: forward and backward iterations. It has the advantage of finding the joints angle all simultaneously. This algorithm starts from the end effector position to the base of the manipulator which is called: the forward iterative method. In the backward iterative method, it works backward in a similar way to complete the whole iteration. Uniquely, this method calculates the position of the joints on a line instead of calculating the angles. Thus, FABRIK has the advantage in saving computation time.

Once the final posture and the location of each joint is calculated, a free obstacle-path is generated from the initial posture to the final posture. Then, this path is fed into a controller. An example of a control system is the PID independent joint control. In this scheme, each joint is controlled by a PID controller, and considered as a linear system. The non-linear coupling effects disturbances are treated as disturbances. In [7], the control system is based on a position-velocity-acceleration feedforward controller and a PID controller for each joint. The independent joint controller recompenses the static and dynamic couplings that exist between the joint while guaranteeing a perfect trajectory tracking.

A more advanced control system is presented in [8]; a nonlinear adaptive output feedback control system for flexible-joint space manipulators. To ensure stabilizing the boundary-layer model, a linear correction term is added to a decentralized adaptive control-based adaptation control system. Moreover, to guarantee the stability of the quasi-steady-state model, a strictly passivity-based approach is utilized to achieve the required stability level.

Fractional-order PID controllers which was firstly invented by Podlubny has been a main area of research in the robotics control field [9]. The well-known control system characterizes with the addition of the external fractional regulation parameters λ , μ , and thus it is sometimes expressed as $PI\lambda D\mu$. This addition broadens the control scale of controllers; it turns them to be more flexible and robust to the

controlled system, which usually enhances the system's performance at the same time.

In [10], the authors implement a control system that is a combination of a fuzzy logic controller, and a fractional PID controller, with an automatic parameter tuning method. The idea was to allow all the five parameters of fractional-order PID controller to vary during operation. This paper proposes a tracking control problem for robotic manipulator based on the FABRIK inverse kinematics technique, and the PID controller.

The inverse kinematics uses the Kinematics equation to estimate the parameters of each degree of freedom in the manipulator. A new heuristic iterative approach has been introduced to solve the problem of Inverse Kinematics called FABRIK method. Our controller system combines the features of FABRIK, and the PID controller to achieve as simple, fast, and roust controlling system.

The paper is organized as follow: system description and FABRIK algorithm in section 2. Control law in section 3. Simulation results in section 4 and the conclusion in section 5.

II. SYSTEM DESCRIPTION AND FABRIK ALGORITHM

Fig.1 presents the block diagram of the system design. The Figure shows the different main parts, namely the FABRIK IK algorithm, path planner, and the PID controller.

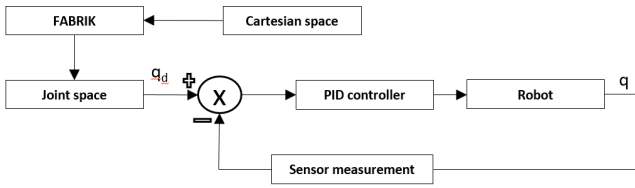


Fig. 1 system design

The 3 DOF manipulator that this paper studies is presented in figure 2. This paper assumed that the mass distribution of each link is to be on the joint ($I=0$). The mass and the link of each link in 1 kg and 1 m respectively.

The DH parameters of the manipulator is shown below:

Joint	variable	α_{i-1}	a_{i-1}	d_i	θ_i
1	θ_1	0	0	L1	θ_1
2	θ_2	90	L2	0	θ_2
3	θ_3	0	L3	0	θ_3

α_{i-1} is the distance from Z_{i-1} to Z_i measured along X_{i-1}

a_{i-1} is the angle from Z_{i-1} to Z_i measured about X_{i-1}

d_i is the distance from X_{i-1} to X_i measured along Z_i

$q\theta_i$ is the angle rotation from X_{i-1} to X_i about Z_i

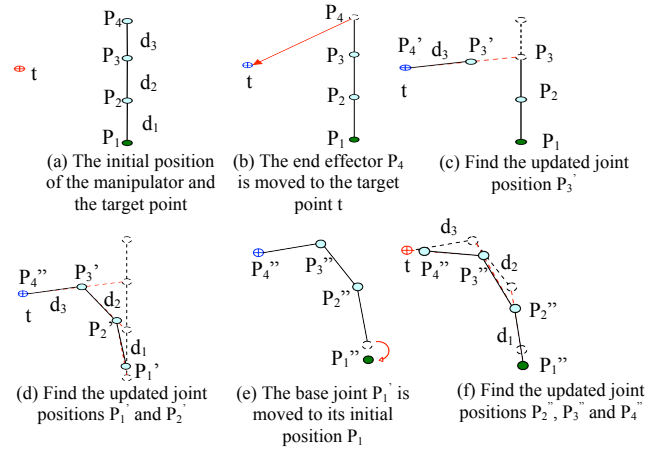


Fig. 2 FABRIK algorithm process

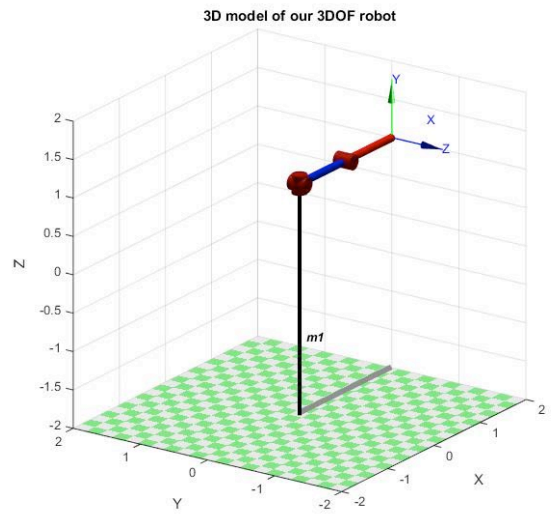


Fig. 3. 3D modeling of the robot using Robotics Toolbox.

The FABRIK inverse kinematics algorithm starts its work from the end effector position all the way to the base of the manipulator which is called the forward iterative method. In the backward iterative method, it works backward in a similar way to complete the whole iteration. Uniquely, this method calculates the position of the joints on a line instead of calculating the angles.

An example of FABRIK for one target and 3 DOF manipulator is illustrated in fig.2. Let P_i ($i=1\sim4$) be the joint position of the manipulator, and P_1 is the base joint and P_4 is the end effector. Let t be the target position and P_i ($i=1\sim4$) be the length of each link. A full iteration of the FABRIK algorithm described as follows:

Firstly, the length of the links $L_i=|P_{i+1}-P_i|$ ($i=1\sim3$) is calculated. Then, check if the target is within the reachable range of the end effector. If $|d_1+ d_2+ d_3|\geq|t- P_1|$, the target position is reachable. Otherwise, it is unreachable. If the target

position is reachable, a full iteration (forward and backward) is performed. The initial position of the robotic manipulator and the target position t are given in Fig.2(a), and Figs. 2(b) ~ (d) show the forward iteration while Figs. 2(e) ~ (f) show the backward iteration. In Fig. 1(b), the end effector P4 moves to the target position t , $P4' = t$. In Fig. 2 (c), the updated joint position P3' is obtained which located on the line d3 that is passing through the points P4' and P3, and $P4' - P3' = d3 = P4 - P3$. For the rest of the joints P1 and P2, the same method is adopted to approach the updated joint positions P1' and P2' (see Fig. 2 (d)). In Fig. 2 (e), the base joint P1' is moved back to its initial position P1, $P1'' = P1$. In Fig. 2 (f), start from the base joint P1 and move backwards to the end effector t , continue repeating the same procedure to obtain updated joint positions P2'', P3'' and P4''. After number of iterations the distance between the end effector should be reduced. This procedure is repeated until the distance between the end effector point and the target position t is less than the tolerance ϵ .

The iterative method of the FABRIK algorithm:

Input: the position of each joint P_i for $i=1, \dots, n$, the position of the target point t and the length of each link $L_i = |P_{i+1} - P_i|$ for $i=1, \dots, n-1$

Output: the new positions P_i of each joint for $i=1, \dots, n$ after k iterations.

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1.1 %find the distance between the base joint  $P_1$  and the
    target point  $t$ 
1.2  $dist = |P_1 - t|$ 
1.3 %check the availability of reaching the target
1.4 If  $dist > L_1 + L_2 + \dots + L_{n-1}$  then
1.5 %the target cannot be reached
1.6 for  $i=1, \dots, n-1$  do
1.7 %find the distance  $r_i$  between the target point  $t$  and  $P_i$ 
1.8  $r_i = |t - P_i|$ 
1.9  $\lambda_i = L_i / r_i$ 
1.10 %Find the new joint position  $P_i'$ 
1.11  $P_{i+1} = (1 - \lambda_i) P_i + \lambda_i t$ 
1.12 end
1.13 end
1.14 %if the target is reachable;  $b$  is the considered as the
    initial position of the base joint  $P_1$ 
1.15  $b = P_1$ 
1.16 % Check if the distance between the end effector position
     $P_n$  and the target point  $t$  is greater than a tolerance.
1.17  $difA = |P_n - t|$ 
1.18 while  $difA > tolerance$  do
1.19 %FORWARD ITERATION
1.20 %Set the end effector  $P_n$  as target  $t$ 
1.21  $P_n = t$ 
1.22 for  $i=n-1, \dots, 1$  do
1.23 % find the distance  $r_i$  between the new joint position  $P_{i+1}$ 
    and the joint  $P_i$ 
1.24  $r_i = |P_{i+1} - P_i|$ 

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1.25  $\lambda_i = L_i / r_i$ 
1.26 % find the updated joint positions  $P_i'$ 
1.27  $P_i = (1 - \lambda_i) P_{i+1} + \lambda_i P_i$ 
1.28 end
1.29 %BACKWARD ITERATION
1.30 %Set the base joint  $P_1$  its initial position
1.31  $P_1 = b$ 
1.32 for  $i=1, \dots, n-1$  do
1.33 % find the distance  $r_i$  between the new joint position  $P_i$ 
    and the joint  $P_{i+1}$ 
1.34  $r_i = |P_{i+1} - P_i|$ 
1.35  $\lambda_i = L_i / r_i$ 
1.36 % find the new joint positions  $P_i'$ 
1.37  $P_i = (1 - \lambda_i) P_i + \lambda_i P_{i+1}$ 
1.38 end
1.39  $difA = |P_n - t|$ 
1.40 end
1.41 end

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III. CONTROL LAW

This section presents the control law that has been used for the 3 DOF manipulator. We used PID controller. PID stands for Proportional K_p , Integral K_i Derivative K_d . By trial and error, we adjust the values of K_p , K_i and K_d to decrease the error. K_p is used to decrease the rise time, K_d is used to reduce the overshoot and settling time and K_i to eliminate the steady-state error.

The general equation of the PID controller is:

$$u = K_p e + K_i \int e dt + \frac{K_d de}{dt}$$

$$u(s) = \left(K_p + \frac{K_i}{s} + K_d s \right) E(s), \text{ where } E(s) \text{ is the error}$$

IV. SIMULATION RESULTS

The system which was shown in Figures 1,2 and 3 was used to demonstrate the performance of the proposed control strategy.

Figure 4 shows the initial configuration of the manipulator. The coordinate of the 3 joints in the cartesian space are as follow: [1,3], [2.5,3], [4,3], [5.5,3] which corresponds to [0,0,0] degrees in the joint space. The goal position was set to be [3,4]. Using the FABRIK algorithm, the end-effector has successfully reached the target [2.9994,4.0006], 0.0006. The final position of all joint was [1,3], [2.5,2.9937], [3.9956,2.8792], [2.9994,4.0006], with an error of 0.0063,0.0044,0.0006 for each joint respectively. Figure 6 shows the number of iterations needed to reach the target against the distance between target and end effector as this changes over time.

Then a straight-line path is generated from the initial positions to the final positions using linear interpolation. This path is fed into the PID controller using 10 as P, 3 as I, and 100 as D. Regarding the tracking results, the controller was able to perfectly track the path with minimal errors. Figures 7-8 shows the tracking in the x and y. Figures 9-10 illustrates that the tracking error is almost zero all the time, which indicates that the PID parameters were perfectly selected to obtain this result.

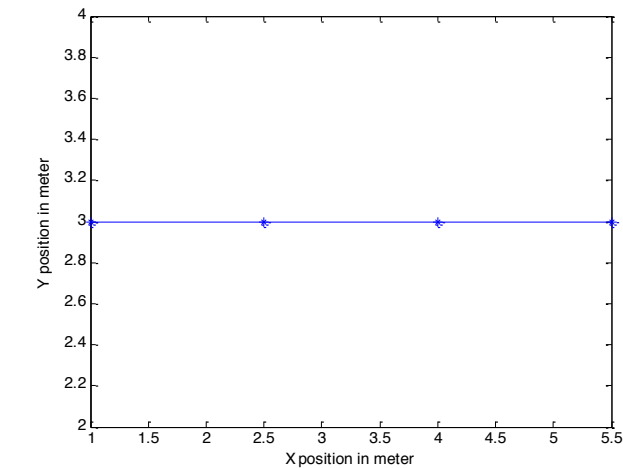


Fig. 4 initial configuration

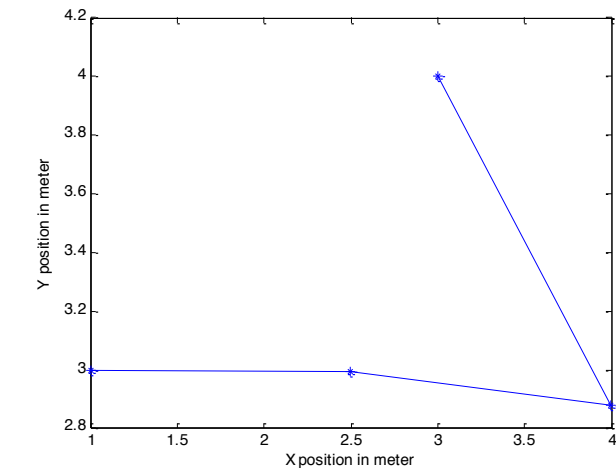


Fig. 5 Final configuration calculated by FABRIK IK

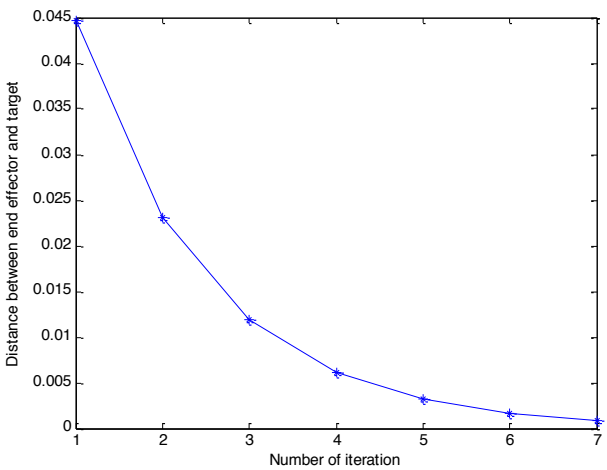


Fig. 6 The number of iterations needed to reach the target against the distance between target and end effector in meter as this changes over time.

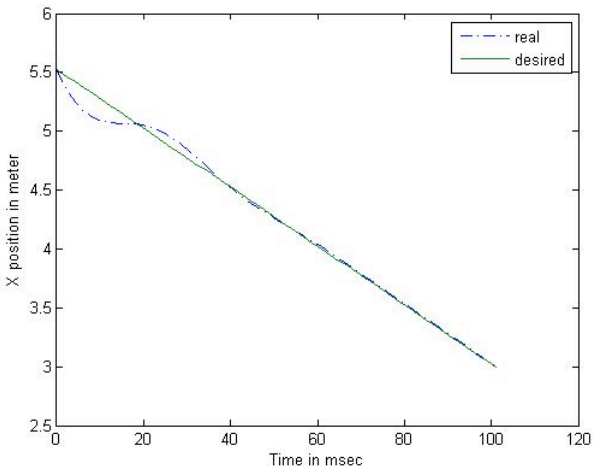


Fig. 7 Tracking control for X position

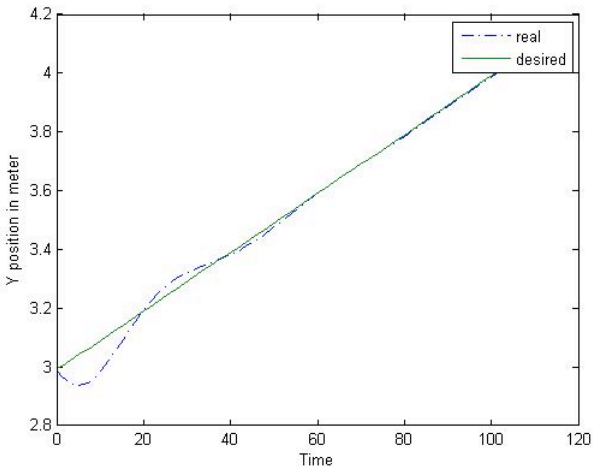


Fig. 8 Tracking control for Y position

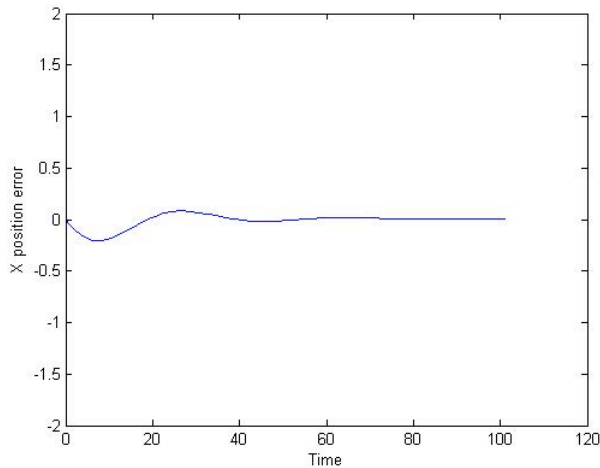


Fig. 9 Error for X position

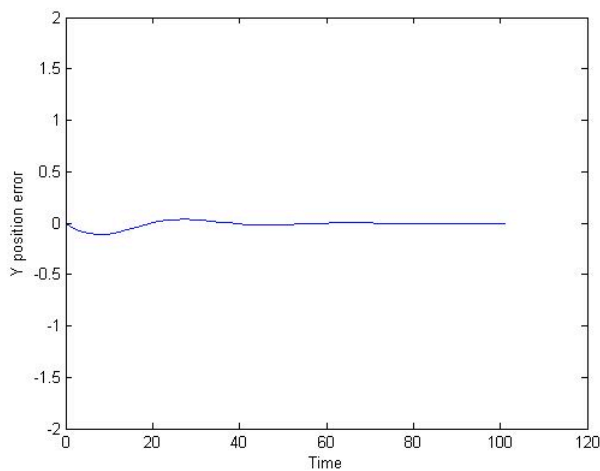


Fig. 10 Error for Y position

CONCLUSION

This paper presented a PID tracking control for manipulator based on FABRIK IK algorithm. This algorithm starts its work from the end effector position all the way to the base of the

manipulator which is called: the forward iterative method. In the backward iterative method, it works backward in a similar way to complete the whole iteration. The simulation results show a good tracking for the manipulator in the joint space. For future work we can use this method furthermore in path planning problems and obstacle avoidance.

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